

Manuel Bonilla López, PhD.

Scientist-Engineer

Data, Systems and Engineering Transformation

321-381-7649 | mbonilla1888@gmail.com | Palm Bay, FL | Active Secret DOD Clearance
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----- PROFESSIONAL SUMMARY -----

Lead-level scientist-engineer with a Ph.D. in Physics and 16+ years of experience driving transformation across advanced manufacturing organizations. Proven leader in translating complex engineering and manufacturing environments into scalable systems, standards, and governance models that improve productivity, reliability, and executive decision-making.

Combines deep hands-on experience as an experimental scientist and process engineer with enterprise-scale systems leadership, enabling effective ownership of manufacturing execution systems, analytics frameworks, and cross-site digital initiatives. Recognized for architecting reference solutions adopted across multiple sites and disciplines, spanning semiconductors and optics manufacturing. Trusted technical leader and internal consultant, partnering directly with senior leadership to define roadmaps, and deliver measurable operational impact through data-driven execution.

----- AREAS OF EXPERTISE -----

LEADERSHIP & OPS.	Digital transformation Multi-site digital governance Change Review Board Capacity & headcount planning Budget and license optimization Statistical Process Control (SPC)
MANUFACTURING	Manufacturing Execution System (MES) Engineering workflows & process routes Tool and product tracking Visual Task Boards Environmental & health monitoring Design of Experiments (DOE) Root Cause and Corrective Action (RCCA)
DEPOSITION SYSTEMS	Molecular Beam Epitaxy (MBE) Atomic Layer Deposition (ALD) E-beam evaporation Sputtering Thermal Evaporation Plasma-Enhanced Chemical Vapor Deposition (PECVD)
AUTOMATION & DATA	SharePoint Power BI dashboards MS Teams alerts Database management Data models and metrics governance GitHub
PROGRAMMING	Python R Microsoft VBA SQL HTML/JSON/CSS MATLAB JMP Origin Lab
OPTICAL DESIGN	FilmStar OpenFilters TFCalc McLeod
METROLOGY	Spectrophotometer Ellipsometers Scanning Tunneling Microscopy (STM) X-RAY Spectroscopy (XPS) Physical Property Measurement System (PPMS) Atomic Force Microscope (AFM) Scanning Electron Microscope (SEM) Electrical characterization

----- PROFESSIONAL EXPERIENCE -----

L3Harris — Sr. MANUFACTURING ENGINEER (MES AND AUTOMATION LEAD) 03/24 – Present
Operating at lead/principal scope; reporting directly to Director of Operations.

- **Expanded Scope: Enterprise & Multi-Site MES Leadership (Continuous Role)** | Multi-Site 11/25 – Present
- Selected as reference MES architect for enterprise expansion across AZ, OH, NH, and FL sites.
 - Leading cross-sector MES architecture, governance, and deployment supporting semiconductor and night-vision/optics manufacturing organizations.
 - Conducting executive demos, scope definition, and roadmap alignment with site leadership and future MES administrators.
 - Training future site MES managers, project leads, and administrators to ensure sustainable adoption.
 - Driving enterprise MES licensing alignment to reduce redundant six-figure annual site costs and enable shared corporate contracts.

- Architected and deployed the Eyelit MES as the foundational manufacturing system for the Microelectronics Department (~200 users), serving as the sole MES owner and systems authority, accountable for governance, configuration, enhancements, and sustainment.
- Established department-wide MES engineering and manufacturing governance, defining naming conventions, operational hierarchies, ownership rules, and required data collection standards across engineering, quality, operations, and maintenance.
- Transformed informal Word-based procedures into structured digital travelers integrating recipes, parameters, inspections, and decision logic to enable traceability and execution consistency.
- Designed and institutionalized the department's first end-to-end SPC and analytics framework, authoring plans, standards and metrics governance.
- Built integrated SharePoint and Power BI operational platforms enabling real-time visibility into tool status.
- Implemented environmental health monitoring strategy, including RH/temperature logger deployment, ISO-based alarm limits, and automated Python + Task Scheduler pipelines feeding live dashboards and Teams notifications.
- Served on the Change Review Board, evaluating engineering and manufacturing process changes.
- Developed the facilities manpower and workload model for the department by analyzing all building-level facilities assets. Reviewed manuals, extracted required preventive maintenance tasks, and validated labor hours and frequencies through direct collaboration with shared facilities personnel. Built a year-long task calendar incorporating a two-shift operating model, no-weekend coverage, vacation and non-productive time assumptions, and a 75% utilization standard directly enabling approval and hiring of dedicated facilities personnel assigned exclusively to the department's buildings.
- Awarded the Rising Star Award by sector-level operations leadership for consistently exceeding performance expectations and delivering high-impact manufacturing technology solutions, including successful implementation of the Eyelit MES and automated visualization platforms. Recognition cited innovation, cross-functional collaboration, and long-term project adoption and sustainability across the organization.

LOCKHEED MARTIN — Sr. OPTICAL MANUFACTURING PROCESS ENGINEER | Orlando, FL 01/22 – 06/23

Led the oxide deposition process team supporting thin-film optical coating programs, overseeing deposition systems, yield improvement initiatives, and cross-functional coordination between operations, engineering, and vendors.

- Led the oxide thin-film group producing antireflective and high-reflective optical filters across the visible and infrared spectra.
- Tool owner for PVD and CVD deposition systems from Tecport Optics and Bühler Leybold, including e-beam, sputtering, thermal evaporation, and PECVD chambers.
- Implemented structured RCCA methodologies that significantly reduced defective runs and enabled production delivery up to one month ahead of schedule for select programs for the first time in the department's history.
- Resolved critical issues involving coating delamination, uniformity, durability, spectral performance, chamber leaks, temperature calibration, and automated cleaning systems.
- Recovered previously out-of-production tools, including an environmental test chamber and a PECVD DLC system, through troubleshooting of water filtration, feedthroughs, and shutdown logic.
- Identified systemic vendor quality gaps contributing to manufacturing failures and led corrective actions aligning supplier capabilities with program requirements.
- Awarded the Performance Management Team Advocate Award for serving as the primary interface between operators, engineering, and management by leading PMT weekly meetings bringing visibility to operator issues, coordinating and assigning corrective actions across teams, and directly resolving a significant portion of production challenges through hands-on troubleshooting and process improvements.

INTEL CORPORATION — SEMICONDUCTOR PROCESS ENGINEER | Hillsboro, OR

06/20 – 05/21

Front-end manufacturing tool owner supporting high-volume semiconductor production at Intel's D1 fab.

- Tool owner for Atomic Layer Deposition (ALD) systems depositing high-k dielectric films for FinFET transistor gates.
- Owned two manufacturing tools comprising four deposition sub-chambers, overseeing film thickness, defect density, and data trending.
- Reduced tool downtime and improved availability by optimizing preventive maintenance strategies using DOEs, pump life analysis, and chamber pressure diagnostics.
- Coordinated with equipment vendors to sustain high-volume production and resolve equipment performance issues.
- Led technology transitions of two previously obsolete tools into active production, restoring manufacturing capability and supporting long-term production continuity.

----- RESEARCH & TECHNICAL EXPERIENCE -----

MATERIALS SCIENCE & SURFACE PHYSICS RESEARCH ASSOCIATE | USF Tampa, FL 08/13 – 08/20

Led hands-on experimental research in thin-film deposition, ultra-high vacuum systems, and advanced materials characterization, with full ownership of process development, equipment operation, and data analysis.

- Specialized in surface science and two-dimensional materials with a focus on physical vapor deposition using MBE.
- Synthesized and characterized monolayer Vanadium Selenide (VSe₂), providing the first published experimental evidence of room-temperature two-dimensional ferromagnetism, foundational to the modern 2D magnetism field. Developed and optimized deposition recipes for monolayer TiSe₂, MoSe₂, WSe₂, and TaSe₂ to study phase behavior, charge density waves, magnetic states, excitonic properties, band structure, doping, and surface morphology.
- Installed, configured, and maintained ultra-high vacuum systems.
- Led installation and commissioning of a shared departmental MBE deposition system, supporting multi-user research.

MATERIALS SCIENCE RESEARCH ASSOCIATE | UPR Humacao, PR 08/10 – 08/13

Conducted hands-on research in gas-sensing device fabrication, integrating polymer nanofiber deposition, microfabrication techniques, and electrical characterization to develop functional sensor devices for defense and smart-textile applications.

- Fabricated gas-sensing diodes and transistors by electrospinning polymer nanofibers directly onto silicon wafers, forming active sensing layers for chemical detection.
- Performed metal electrode deposition via thermal evaporation to define device contacts and circuit structures.
- Executed electrical characterization including I–V measurements and gas response testing to evaluate device sensitivity, stability, and performance.
- Conducted structural and morphological analysis using SEM to validate nanofiber formation and device architecture.
- Developed and optimized polymer formulations and nanofiber recipes to tune chemical composition and improve gas-sensing response characteristics.
- Supported early research toward smart-textile concepts integrating sensing devices into fabrics for environmental and defense-related gas detection.

----- EDUCATION -----

Ph.D., Physics, University of South Florida (USF)

M.Sc., Physics, University of South Florida (USF)

B.Sc., Physics Applied to Electrical Engineering, University of Puerto Rico (UPR)

Completed near-minor-level coursework in Computational Mathematics, University of Puerto Rico (UPR)

----- CERTIFICATIONS & PUBLICATIONS -----

Google Data Analytics Professional Certificate | Google via Coursera

Leading People and Teams Certificate — In Progress | University of Michigan via Coursera

CS50 (Multiple Computer Science Courses) — In Progress | Harvard University via edX

Personal Data Analytics Project (Finance) | [Kaggle Portfolio](#)

10 peer-reviewed publications with 2,300+ citations | [Google Scholar Profile](#)

Research highlighted by Nature Nanotechnology editorial: “[2D Magnetism Gets Hot](#)” | Nature Nanotechnology

----- GUEST SCIENTIST -----

GUEST SCIENTIST — National Research Laboratories (Particle Accelerator Facilities)

- SLAC National Accelerator Laboratory | California, USA
- SOLEIL Synchrotron | Paris, France.